Numerical processing of financial data

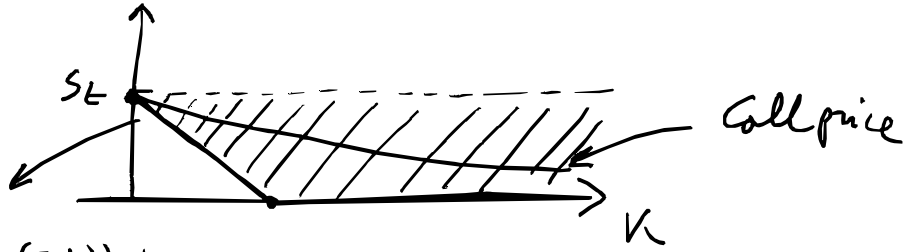
Session 6
13/02/2024

lecture: implied volatility

- **No arbitrage bounds on option prices:**

Let $Call_t(T,K)$ and $Put_t(T,K)$ be call and put prices (for ex., market quotes)
If there is no arbitrage on the market, the following bounds hold:

$$\boxed{\begin{aligned} \left(S_t - Ke^{-r(T-t)}\right)^+ &\le Call_t(T,K) < S_t \\ \left(Ke^{-r(T-t)} - S_t\right)^+ &\le Put_t(T,K) < Ke^{-r(T-t)} \end{aligned}}$$



$\left(S_t - Ke^{-r(T-t)}\right)^+$

<u>proof</u>: at maturity T, we have, for calls

$$S_T - K \leq (S_T - K)^+ < S_T \qquad \forall K > 0$$

Then, if no arbitrage

$$\text{price}_t(S_T - K) \leq \text{price}_t\left((S_T - K)^+\right) < \text{price}_t(S_T)$$

$$\Downarrow$$

$$\underbrace{S_t - e^{-r(T-t)}K}_{} \leq \text{Call}_t(T, K) < S_t \qquad \forall K > 0, \ \forall T, \forall t \leq T$$

We can improve this lower bound

to $\max\left(S_t - e^{-r(T-t)}K, 0\right) = \left(S_t - K e^{-r(T-t)}\right)^+$

- For the put option: same argument (exercise) $\square$

▷ Another fact:

Black Scholes optian prices tend to their lower and upper bounds as $\sigma \to 0$ and $\sigma \to \infty$.

- For the BS put price, we have:

$$\mathrm{Put}_t^{BS}(T,K)\Big|_{\sigma=0} = E\left[e^{-r(T-t)}\left(K - S\, e^{\sigma\sqrt{T-t}\,G + (r-\frac{\sigma^2}{2})(T-t)}\right)^+\right]_{\sigma=0}$$

$$= e^{-r(T-t)}\left(K - S\, e^{r(T-t)}\right)^+$$

$$= \left(e^{-r(T-t)}K - S\right)^+ = \text{lower bound}$$

and

$$\lim_{\sigma\to\infty} \mathrm{Put}_t^{BS}(T,K) = \lim_{\sigma\to\infty}\left(Ke^{-r(T-t)}N(-d_2) - SN(-d_1)\right)$$

$$= Ke^{-r(T-t)}\underbrace{N(+\infty)}_{=1} - S\underbrace{N(-\infty)}_{0}$$

$$= Ke^{-r(T-t)} = \text{upper bound}$$

we use:

$$d_1 = \frac{\ln\frac{Se^{r(T-t)}}{K}}{\sigma\sqrt{T-t}} + \frac{\sigma\sqrt{T-t}}{2} \xrightarrow[\sigma\to\infty]{} +\infty$$

$$d_2 = \text{—}\,''\,\text{—} - \frac{\sigma\sqrt{T-t}}{2} \xrightarrow[\sigma\to\infty]{} -\infty$$

▷ <u>Options implied volatility</u>:

Let t fixed. Consider a function $(t = t_0 = 0 = \text{today})$

$$\text{Put}_t(T, K)$$

representing a put price.

<u>Definition</u>: the implied volatility of the put price $\text{Put}_t(\cdot,\cdot)$ is the function

$$\sigma_t(T, K)$$

defined implicitly by the equation:

$$\boxed{\underbrace{\text{Put}_t(T,K)}_{\text{the given put price}} = \underbrace{\text{Put}_t^{BS}(T,K,\sigma)}_{\text{BS put price with same maturity and strike}}} \quad (*)$$

- The solution to equation $(*)$ is a function of T, K that we denote $\sigma_t(T,K)$.
- Note this is an application of the <u>implicit function theorem</u>:

given $f(x,y)$ and $g(x)$
look for $y(x)$ such that

$$f(x, y(x)) = g(x)$$

Condition: $\partial_y f(x,y) \neq 0$

- Of course, the implied volatility is nothing but the <u>inverse</u> of the BS price w.r.t. σ:

$$\sigma_t(T,K) = \left(\mathrm{Put}_t^{BS}(T,K,\cdot)\right)^{-1}\left(\mathrm{Put}_t(T,K)\right)$$

Is this inverse (or implicit) function well defined?

Does equation ⊛ have a unique solution, $\forall (T,K)$?

▷ <u>Well-posedness</u> of ⊛:

- $\sigma(T,K)$ is well-defined as soon as the given price $\mathrm{Put}_t(T,K)$ satisfies the no-arbitrage bounds:

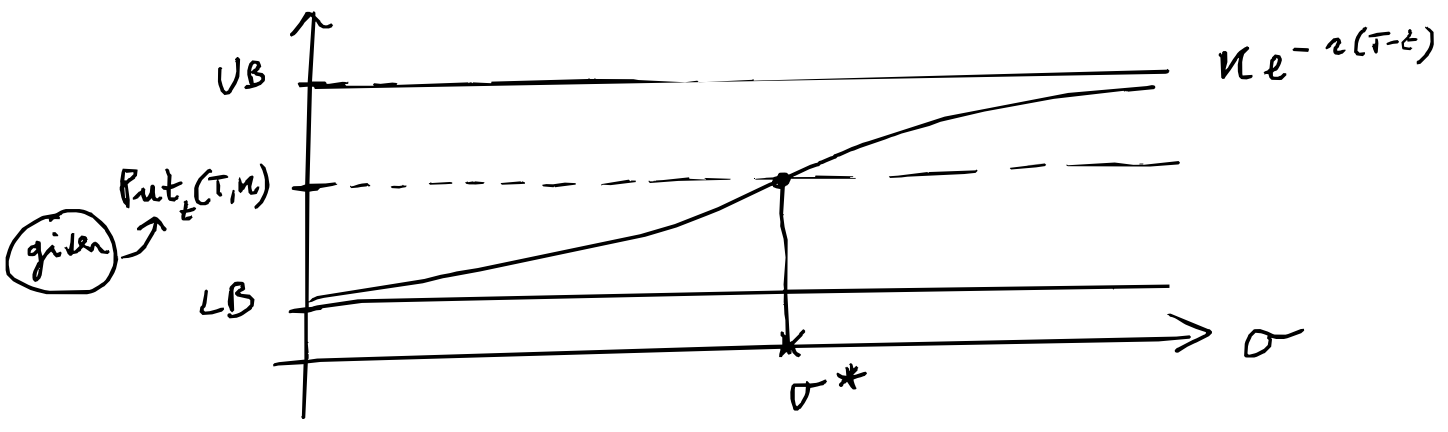
$$\left(Ke^{-r(T-t)} - S_t\right)^+ \leq \mathrm{Put}_t(T,K) < Ke^{-r(T-t)}$$

- On the other side of the equation, we have:

$$\lim_{\sigma \to 0} \mathrm{Put}_t^{BS}(T,K) = \text{lower bound} = LB$$

$$\lim_{\sigma \to \infty} \mathrm{Put}_t^{BS}(T,K) = \text{upper bound} = UB$$

hence a <u>solution</u> for σ exists in ⊛



for fixed T, fixed K

→ In other words, the function $\sigma \mapsto \mathrm{Put}_t^{BS}(T,K,\sigma)$ is surjective onto the interval $[LB, UB)$.

- In addition, we have (BS Vega)

$$\partial_\sigma \, \mathrm{Put}^{BS}(T,K,\sigma) = S\sqrt{T-t}\, n(d_1) > 0 \quad \text{always}$$

Therefore, the <u>solution σ^* is unique</u>.

$\rightarrow$ In other words, the function $\sigma \mapsto \mathrm{Put}_t^{BS}(T,K,\sigma)$ is strictly increasing ($\forall T, \forall K, \forall t < T$).

▷ <u>Note</u>:
even if $\mathrm{Put}^{BS}(T,K,\sigma)$ has an explicit formula, in general the implied volatility $\sigma(T,K)$ does not.

↳ the equation $\mathrm{Put}_t^{BS}(T,K,\sigma) = \mathrm{Put}_t(T,K)$ has to be solved numerically for σ.

<u>Numerical methods</u>:

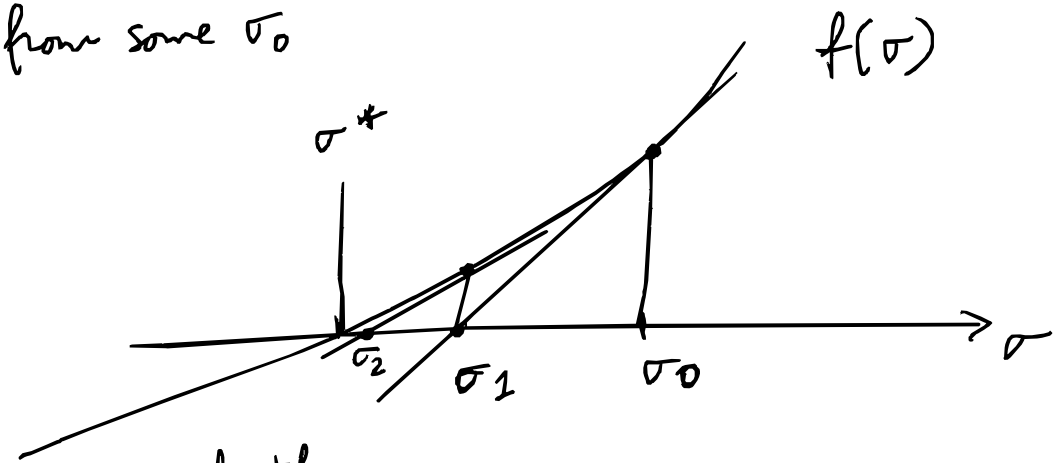
we have to look for the (unique) zero of the function

$$f(\sigma) = \mathrm{Put}_t^{BS}(T,K,\sigma) - \mathrm{Put}_t(T,K) = 0$$

<u>Possible methods</u>:

- Newton's method (gradient descent, in dim 1)
- Bisection method

· <u>Newton's method</u> (in dim 1) : iterative method that starts from some σ_0



Step 1: we look for σ such that

$$0 = \text{line}(\sigma) = f(\sigma_0) + f'(\sigma_0)(\sigma - \sigma_0)$$

$$\Rightarrow \sigma_1 := \sigma_0 - \frac{f(\sigma_0)}{f'(\sigma_0)}$$

Step ℓ: iteration ℓ

$$\boxed{\sigma_{\ell+1} = \sigma_\ell - \frac{f(\sigma_\ell)}{f'(\sigma_\ell)}}$$

where we have

$$f(\sigma) = \text{Put}_t^{BS}(T, K, \sigma) - \text{Put}_t(T, K)$$

$$f'(\sigma) = \text{Vega}_t^{BS}(T, K, \sigma)$$

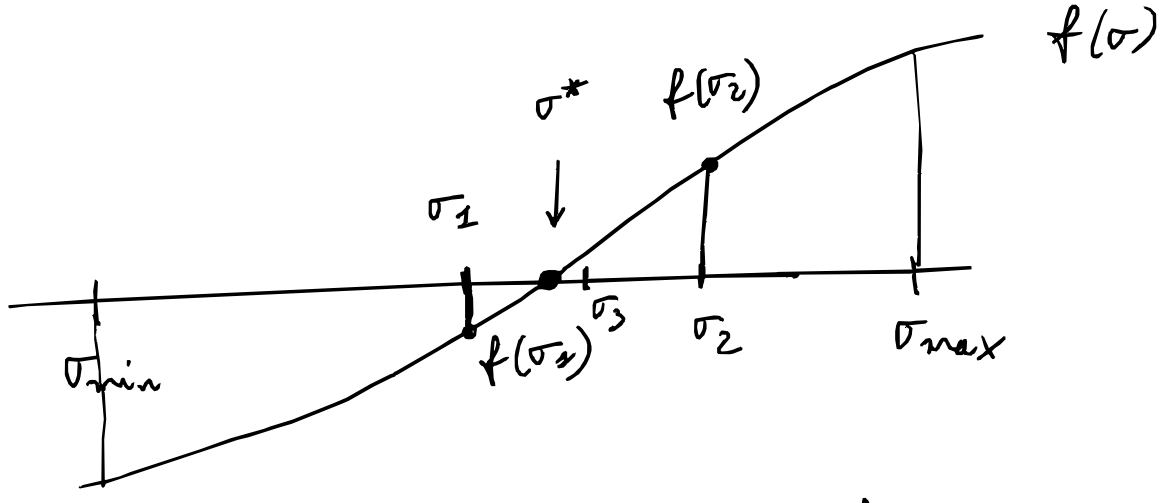
▷ A first order method:

we use the knowledge of f'.

▷ <u>Bisection method</u>:

- Starts with σ_{min} such that $f(\sigma_{min}) < 0$

  σ_{max} such that $f(\sigma_{max}) > 0$



- Evaluate $\sigma_1 = \dfrac{\sigma_{min} + \sigma_{max}}{2}$ and $f(\sigma_1)$
- According to the sign of $f(\sigma_1)$, select one of the two sub-intervals
- Evaluate $\sigma_2 = \dfrac{\sigma_1 + \sigma_{max}}{2}$
- Iterate until desired precision: stopping criterion can be

$$|f(\sigma_\ell)| < \varepsilon$$

$$\text{or } |\sigma_{\ell+1} - \sigma_\ell| < \varepsilon$$

▷ A <u>zeroth-order method</u>: only evaluates f, not f'.

→ Typically, bisection method is slower (larger number of iterations for the same precision ε), but more stable than Newton's method.

Remarks:

- We can define implied volatility from call options via the equation

$$\underset{\substack{\uparrow \\ \text{given call price}}}{\mathrm{Call}_t(T,K)} = \mathrm{Call}_t^{BS}(T,K,\sigma)$$

$\rightarrow$ defines $\sigma_t^{Call}(T,K)$

$\rightarrow$ Do we have $\boxed{\sigma_t^{Put}(T,K) = \sigma_t^{Call}(T,K)}$?

Yes, by put-call parity (exercise).

- Obviously, if we start from a BS put or call price with some parameter $\bar{\sigma}$:

$$\mathrm{Put}(T,K) = \mathrm{Put}^{BS}(T,K,\bar{\sigma})$$

or $\mathrm{Call}(T,K) = \mathrm{Call}^{BS}(T,K,\bar{\sigma})$

then of course, we get

$$\boxed{\sigma(T,K) = \bar{\sigma}} \qquad \forall (T,K)$$

$\rightarrow$ we simply find the parameter $\bar{\sigma}$ we have used in the BS formula.

- As we will see, for <u>market prices</u>, in general we have

$$\boxed{\sigma(T,K) \neq \text{constant}}$$

▷ The function $K \mapsto \sigma(T,K)$ for fixed T is called the <u>implied vol smile</u>

▷ The whole function $(T,K) \mapsto \sigma(T,K)$ is the <u>implied volatility surface</u>

- The fact the $\sigma(T,K)$ is not constant for all T,K means that market prices of options are not quoted by market participants with the BS formula with constant parameter σ.